

TSIA2 Math Practice Test 9

PART A — QUESTIONS

TSIA2-M9-001

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A student's total fees for one semester are \$1,840. A grant covers 65% of the fees. How much of the fees must the student still pay?

- A) \$598
- B) \$644
- C) \$1,196
- D) \$2,505

TSIA2-M9-002

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve for x : $5(x + 2) - 3x = 26$

- A) 6
- B) 8
- C) 10
- D) 18

TSIA2-M9-003

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular banner is 9 feet long and 4 feet wide. A ribbon is placed around the entire perimeter of the banner. How many feet of ribbon are needed?

- A) 13 feet
- B) 26 feet
- C) 36 feet
- D) 72 feet

TSIA2-M9-004

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The number of calls received at a help desk during five hours was 12, 18, 15, 21, and 19. What is the mean number of calls received per hour?

- A) 15
- B) 16
- C) 17
- D) 18

TSIA2-M9-005

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A water tank is $\frac{3}{8}$ full. If the tank currently contains 24 gallons of water, what is the total capacity of the tank?

- A) 48 gallons
- B) 56 gallons
- C) 64 gallons
- D) 72 gallons

TSIA2-M9-006

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $8m - 3(2m - 5) + 4$?

- A) $2m - 11$
- B) $2m + 19$
- C) $14m - 11$
- D) $14m + 19$

TSIA2-M9-007

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A circular mirror has a diameter of 22 inches. What is the radius of the mirror?

- A) 11 inches
- B) 22 inches
- C) 44 inches
- D) 121 inches

TSIA2-M9-008

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A box contains 6 black clips, 9 silver clips, and 5 gold clips. One clip is selected at random. What is the probability that the clip is silver?

- A) $\frac{3}{20}$
- B) $\frac{9}{20}$
- C) $\frac{11}{20}$
- D) $\frac{3}{5}$

TSIA2-M9-009

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

Enrollment in an online course increased from 148 students to 220 students over 8 weeks. What was the average increase in enrollment per week?

- A) 8 students per week
- B) 9 students per week
- C) 18 students per week
- D) 72 students per week

TSIA2-M9-010

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A linear function has a slope of -4 and a y-intercept of 9. Which equation represents the function?

- A) $y = 9x - 4$
- B) $y = -4x + 9$
- C) $y = 4x - 9$
- D) $y = -9x + 4$

TSIA2-M9-011

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular prism has dimensions 7 inches, 5 inches, and 3 inches. What is its volume?

- A) 15 cubic inches
- B) 35 cubic inches
- C) 75 cubic inches
- D) 105 cubic inches

TSIA2-M9-012

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The data set shows the number of pages read by a student on six days:

18, 24, 24, 30, 36, 42

Which statement is true?

- A) The median is 24 pages.
- B) The median is 27 pages.
- C) The mode is 30 pages.
- D) The range is 18 pages.

TSIA2-M9-013

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A store sells 6 markers for \$7.50. At this same rate, how much would 14 markers cost?

- A) \$15.00

- B) \$16.25
- C) \$17.50
- D) \$21.00

TSIA2-M9-014

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

For the function $g(x) = -x^2 + 6x + 3$, what is $g(4)$?

- A) 5
- B) 8
- C) 11
- D) 19

TSIA2-M9-015

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A right triangle has a hypotenuse of 17 feet and one leg of 8 feet. What is the length of the other leg?

- A) 9 feet
- B) 12 feet
- C) 15 feet
- D) 25 feet

TSIA2-M9-016

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A jar contains 10 coins: 4 quarters, 3 dimes, and 3 nickels. Two coins are selected at random without replacement. What is the probability that both coins are quarters?

- A) $\frac{2}{15}$
- B) $\frac{4}{25}$
- C) $\frac{1}{5}$
- D) $\frac{3}{10}$

TSIA2-M9-017

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A tablet screen is advertised as having a diagonal length of 11.8 inches. Which whole number is closest to 11.8^2 ?

- A) 118
- B) 132
- C) 139
- D) 148

TSIA2-M9-018

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $x^2 + 2x - 48$?

- A) $(x + 6)(x - 8)$
- B) $(x - 6)(x + 8)$
- C) $(x + 4)(x - 12)$
- D) $(x - 4)(x + 12)$

TSIA2-M9-019

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A point at $(3, -5)$ is reflected across the y-axis. What are the coordinates of the reflected point?

- A) $(-3, -5)$
- B) $(3, 5)$
- C) $(-3, 5)$
- D) $(5, -3)$

TSIA2-M9-020

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A poll asked 120 students which campus service they used most often. The results are shown below.

Library: 42 students

Tutoring center: 30 students

Fitness center: 18 students

Computer lab: 30 students

Which conclusion is supported by the data?

- A) More than half of the students chose the library.
- B) The tutoring center and computer lab were chosen by the same number of students.
- C) The fitness center was chosen by 25% of the students.
- D) The library was chosen by twice as many students as the computer lab.

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M9-001 — Correct: B — Correct Answer: \$644 — Explanation: If the grant covers 65%, the student pays the remaining 35%. Then $0.35 \times 1,840 = \$644$.

TSIA2-M9-002 — Correct: B — Correct Answer: 8 — Explanation: Distribute and combine like terms: $5x + 10 - 3x = 26$, so $2x + 10 = 26$. Then $2x = 16$, so $x = 8$.

TSIA2-M9-003 — Correct: B — Correct Answer: 26 feet — Explanation: The perimeter of the rectangle is $2(9 + 4) = 26$ feet.

TSIA2-M9-004 — Correct: C — Correct Answer: 17 — Explanation: The total number of calls is $12 + 18 + 15 + 21 + 19 = 85$. Dividing by 5 gives a mean of 17.

TSIA2-M9-005 — Correct: C — Correct Answer: 64 gallons — Explanation: If 24 gallons is $\frac{3}{8}$ of the tank, the total capacity is $24 \div \frac{3}{8} = 24 \times \frac{8}{3} = 64$ gallons.

TSIA2-M9-006 — Correct: B — Correct Answer: $2m + 19$ — Explanation: Distribute and combine like terms: $8m - 6m + 15 + 4 = 2m + 19$.

TSIA2-M9-007 — Correct: A — Correct Answer: 11 inches — Explanation: The radius is half the diameter, so $22 \div 2 = 11$ inches.

TSIA2-M9-008 — Correct: B — Correct Answer: $\frac{9}{20}$ — Explanation: There are $6 + 9 + 5 = 20$ clips total, and 9 are silver. The probability is $\frac{9}{20}$.

TSIA2-M9-009 — Correct: B — Correct Answer: 9 students per week — Explanation: The increase was $220 - 148 = 72$ students. Over 8 weeks, the average increase was $72 \div 8 = 9$ students per week.

TSIA2-M9-010 — Correct: B — Correct Answer: $y = -4x + 9$ — Explanation: In slope-intercept form, $y = mx + b$. With slope -4 and y-intercept 9, the equation is $y = -4x + 9$.

TSIA2-M9-011 — Correct: D — Correct Answer: 105 cubic inches — Explanation: The volume of a rectangular prism is length $\times$ width $\times$ height, so $7 \times 5 \times 3 = 105$ cubic inches.

TSIA2-M9-012 — Correct: B — Correct Answer: The median is 27 pages. — Explanation: With six values, the median is the average of the third and fourth values: $(24 + 30) \div 2 = 27$.

TSIA2-M9-013 — Correct: C — Correct Answer: \$17.50 — Explanation: The price per marker is $\$7.50 \div 6 = \1.25 . For 14 markers, the cost is $14 \times \$1.25 = \17.50 .

TSIA2-M9-014 — Correct: C — Correct Answer: 11 — Explanation: Substitute 4 for x: $g(4) = -4^2 + 6(4) + 3 = -16 + 24 + 3 = 11$.

TSIA2-M9-015 — Correct: C — Correct Answer: 15 feet — Explanation: The missing leg is $\sqrt{(17^2 - 8^2)} = \sqrt{(289 - 64)} = \sqrt{225} = 15$ feet.

TSIA2-M9-016 — Correct: A — Correct Answer: $\frac{2}{15}$ — Explanation: The probability of selecting a quarter first is $\frac{4}{10}$. Without replacement, the probability of selecting a quarter second is $\frac{3}{9}$. The product is $\frac{4}{10} \times \frac{3}{9} = \frac{12}{90} = \frac{2}{15}$.

TSIA2-M9-017 — Correct: C — Correct Answer: 139 — Explanation: $11.8^2 = 139.24$, which is closest to the whole number 139.

TSIA2-M9-018 — Correct: B — Correct Answer: $(x - 6)(x + 8)$ — Explanation: The numbers -6 and 8 multiply to -48 and add to 2 , so $x^2 + 2x - 48 = (x - 6)(x + 8)$.

TSIA2-M9-019 — Correct: A — Correct Answer: $(-3, -5)$ — Explanation: Reflecting across the y-axis changes the sign of the x-coordinate and leaves the y-coordinate the same.

TSIA2-M9-020 — Correct: B — Correct Answer: The tutoring center and computer lab were chosen by the same number of students. — Explanation: The table shows that 30 students chose the tutoring center and 30 students chose the computer lab.